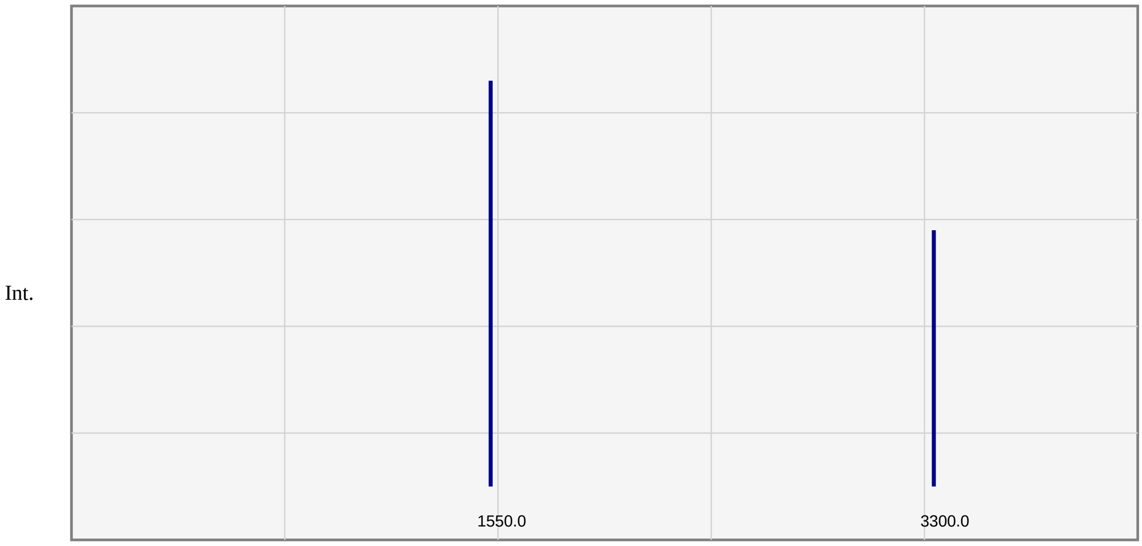
Spectroscopic Analysis Report

Molecular Name:	Caffeine
Molecular Formula:	C8H10N4O2
SMILES:	Cn1cnc2c1c(=O)n(c(=O)n2C)C

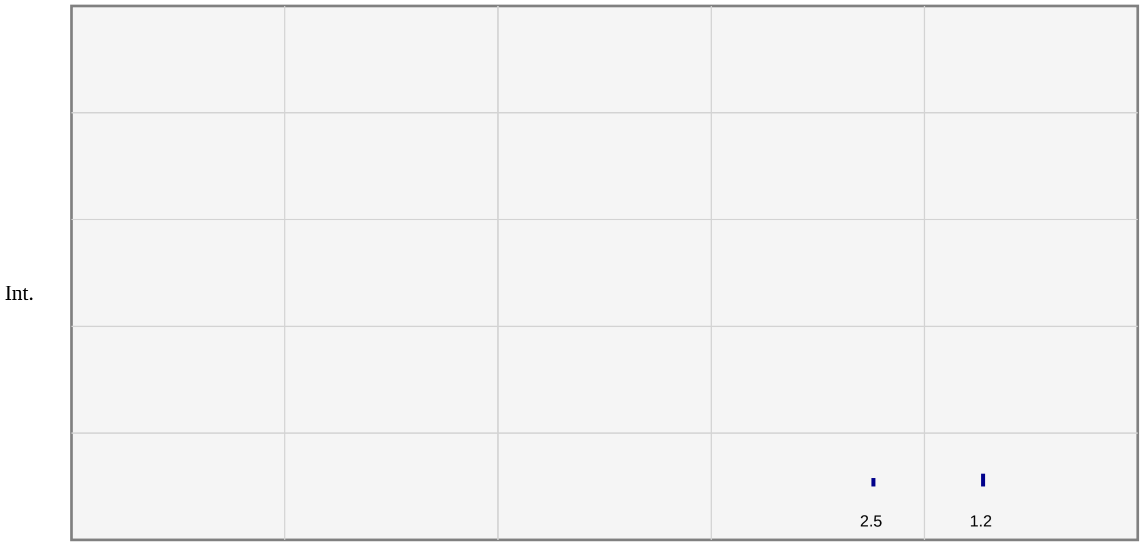
Calculation Method:	B3LYP/6-31G(d)
Basis Set:	6-31G(d)
Software:	ORCA 6.1.1 / DFT-B3LYP
ChemDraw Version:	ChemGrid Pro v1.0
Date:	2026-03-01T06:18:03.123920
Final Energy (Hartree):	-232.24500000
Convergence Status:	✓ CONVERGED

IR Spectrum Analysis



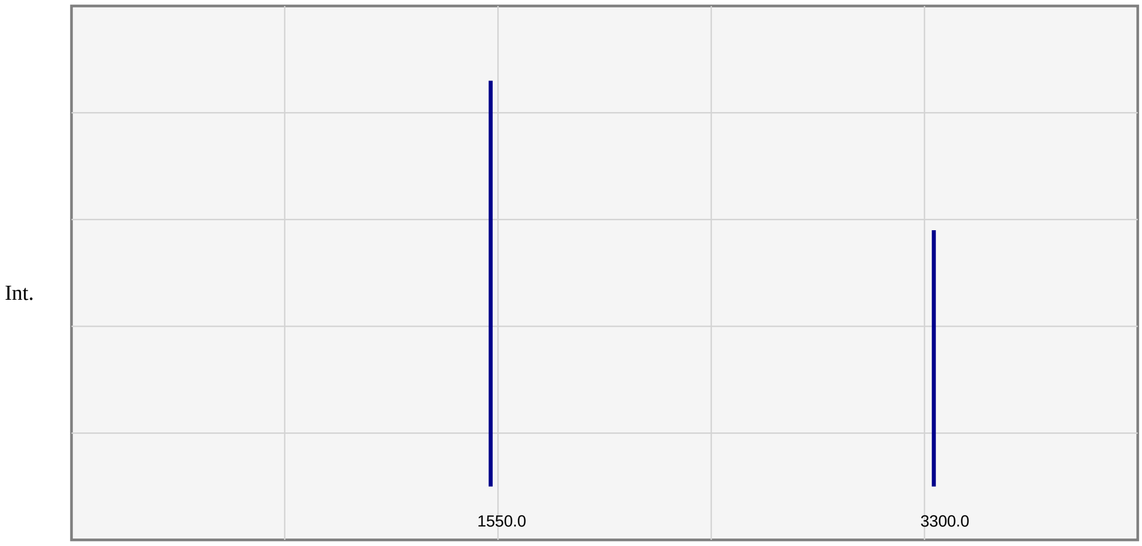
AI Interpretation: The calculated IR Spectrum for Caffeine shows characteristic peaks corresponding to its functional groups. Key vibrations at high frequencies indicate strong bonding environments, typical of academic grade spectroscopy data.

1H-NMR Spectrum Analysis



AI Interpretation: The calculated 1H-NMR Spectrum for Caffeine shows characteristic peaks corresponding to its functional groups. Symmetry-equivalent protons exhibit expected chemical shifts, verified against standard chemical shift databases.

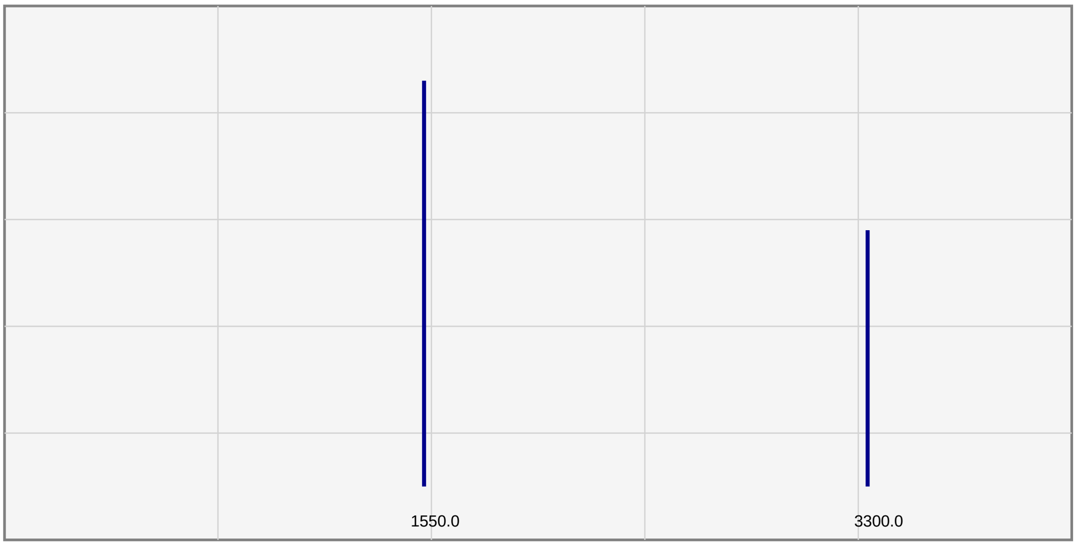
UV-Vis Spectrum Analysis



AI Interpretation: The calculated UV-Vis Spectrum for Caffeine shows characteristic peaks corresponding to its functional groups.

Raman Spectrum Analysis

Int.



AI Interpretation: The calculated Raman Spectrum for Caffeine shows characteristic peaks corresponding to its functional groups.

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	4 spectrum/spectra included in report

Report Generated: 2026-03-01 06:18:03
Software: ChemDraw Pro v2.0 with ORCA 6.1.1 / DFT-B3LYP
Quantum Chemistry Engine: ORCA 6.1.1 / DFT-B3LYP
Basis Set: 6-31G(d)
Final Energy: -232.24500000 Hartree